ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems

Model Answers

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Topic 2: Pass- Level Tutorial Questions

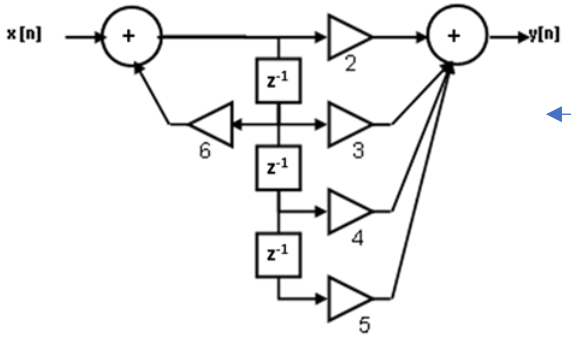
1. What is the equivalent transfer function of the block diagram below?
(Tick the correct answer)

(a) $H(z) = \frac{z(z^{-1}+3)(z^{-1}+4)(z^{-1}+5)}{6-z^{-1}}$

(b) $H(z) = \frac{2+3z^{-1}+4z^{-2}+5z^{-3}}{1-6z^{-1}}$

(c) $H(z) = \frac{1+6z^{-1}}{2+3z^{-1}+4z^{-2}+5z^{-3}}$

(d) $H(z) = \frac{2+3z^{-1}+4z^{-2}+5z^{-3}}{1+6z^{-1}}$



$$\frac{P(z)}{X(z)} = \frac{1}{1-6z^{-1}}$$

$$\frac{Y(z)}{P(z)} = 2 + 3z^{-1} + 4z^{-2} + 5z^{-3}$$

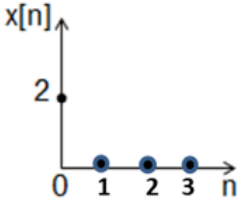
Therefore,

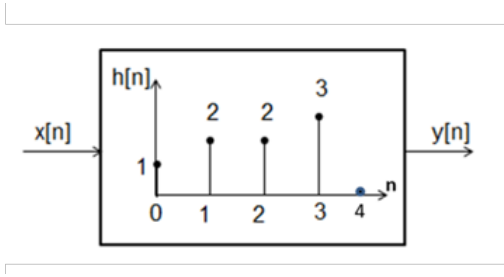
$$\frac{Y(z)}{X(z)} = \frac{Y(z)}{P(z)} \cdot \frac{P(z)}{X(z)}$$

$$= \left(2 + 3z^{-1} + 4z^{-2} + 5z^{-3}\right)\frac{1}{1-6z^{-1}}$$

2. The impulse response ,$h[n]$, of a discrete-time system is given below. The output signal, $y[n]$, for the input signal, $x[n]$ is given by: (Tick the correct answer)

(a) {2 0 0 0 0}
(b) {1 2 2 0 0}
(c) {3 4 5 2 0}
(d) {2 4 4 6 0}
(e) None of the above





$$y[n] = x[n] * h[n] = 2\delta[n] * h[n]$$

where $\delta[n]$is an impulse (unit amplitude).

Therefore,

$$y[n] = 2h[n] = \{2,4,4,6,0\}$$

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3. The impulse response of the system with the transfer function $H(z) = \frac{\alpha+\alpha z^{-1}}{1-\alpha z^{-1}}$ is given by: (Tick the correct answer)

(a) $\alpha^{n-1} + \alpha$ n < 0

(c) α^{n+1} u[n] + α^n u[n-1]

(b) α^n u[n] + α^{n-1} u[n-1]

(d) None of the above

$$H(z) = \frac{\alpha}{1-\alpha z^{-1}} + \frac{\alpha z^{-1}}{1-\alpha z^{-1}}$$

Using the inverse z-transform:

$$h[n] = \alpha(\alpha)^n u[n] + \alpha(\alpha)^{n-1} u[n-1]$$

Therefore, $\boxed{h[n] = \alpha^{n+1} u[n] + \alpha^n u[n-1]}$

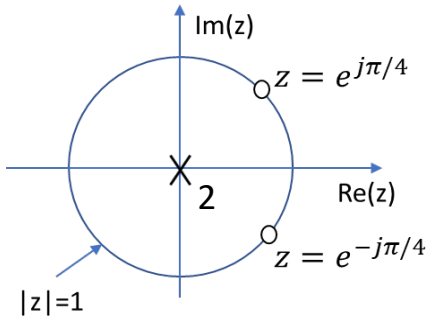
4. The pole-zero pattern of a system has zeros on the unit circle as show below. The system function is given by: (Tick the correct answer)

(a) $H(z) = 1 + \frac{1}{\sqrt{2}} z^{-1} + z^{-2}$

(b) $H(z) = z - \frac{1}{\sqrt{2}} + z^{-1}$

(c) $H(z) = z^2 - 1 + z^{-2}$

(d) $H(z) = 1 - \sqrt{2}\, z^{-1} + z^{-2}$



$$H(z) = \frac{\left(z - 1 \cdot e^{j\frac{\pi}{4}}\right)\left(z - 1 \cdot e^{-j\frac{\pi}{4}}\right)}{z^2} = \frac{z^2 - \left(e^{j\frac{\pi}{4}} + e^{-j\frac{\pi}{4}}\right)z + 1}{z^2}$$

Since $e^{j\frac{\pi}{4}} + e^{-j\frac{\pi}{4}} = 2\cos\left(\frac{\pi}{4}\right) = \sqrt{2},$

we have $H(z) = \frac{z^2 - 2\cos\left(\frac{\pi}{4}\right)z + 1}{z^2} = \frac{z^2 - \sqrt{2}z + 1}{z^2}$

Therefore, $\boxed{H(z) = 1 - \sqrt{2}z^{-1} + z^{-2}}$

5. The DFT of a real signal is given below. The signal energy is: (Tick the correct answer)

(a) 0
(b) 5
(c) 7
(d) None of the above

DFT: [1, $\sqrt{2}$, -1, B, -7, -j2, C, $\sqrt{2}$]

k=0

We use **conjugate symmetry** of the DFT to find Band C:

$$[1 \quad \sqrt{2} \quad -1 \quad B \quad -7 \quad -j2 \quad C \quad \sqrt{2}]$$

Therefore $\boxed{B = j2}$ and $\boxed{C = -1}$

$$X[k] = [1 \quad \sqrt{2} \quad -1 \quad j2 \quad -7 \quad -j2 \quad -1 \quad \sqrt{2}]$$

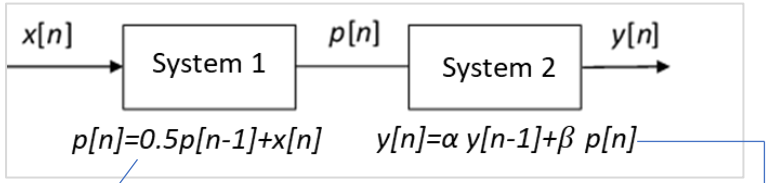
Signal energy: $E = \frac{1}{8}\sum_{k=0}^{7} |X[k]|^2 = \frac{1}{8}\left[1^2 + \left(\sqrt{2}\right)^2 + (-1)^2 + 2^2 + 7^2 + 2^2 + (-1)^2 + \left(\sqrt{2}\right)^2\right] = 8$

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1. Consider the cascade of the following two systems as depicted below. The difference equation relating $x[n]$ and $y[n]$ is given by:

$$y[n] = x[n] + \frac{3}{4}y[n-1] - \frac{1}{8}y[n-2]$$

Determine α and β.



$p[n] = 0.5p[n-1] + x[n]$; Therefore, $\frac{P(z)}{X(z)} = \frac{1}{1-0.5z^{-1}}$

Also, $y[n] = \alpha y[n-1] + \beta p[n]$ Therefore, $\frac{Y(z)}{P(z)} = \frac{\beta}{1-\alpha z^{-1}}$

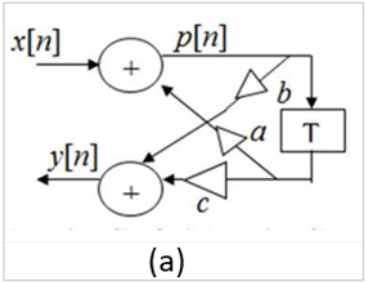
Hence, $\frac{Y(z)}{X(z)} = \frac{Y(z)}{P(z)}\frac{P(z)}{X(z)} = \frac{\beta}{1-\alpha z^{-1}} \cdot \frac{1}{1-0.5z^{-1}}$

Thus, $\boxed{\frac{Y(z)}{X(z)} = \frac{\beta}{1-(\alpha+0.5)z^{-1}+0.5\alpha z^{-2}}}$ (1)

Given, $\boxed{\frac{Y(z)}{X(z)} = \frac{1}{1-\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}}}$ (2)

Equating (1) and (2), $\boxed{\beta = 1}$ and $0.5\alpha = \frac{1}{8}$; Therefore, $\boxed{\alpha = \frac{1}{4}}$

2. For the block diagram realisations given below, develop the relation between y[n] and x[n] in each case.



(a)

$$P(z) = X(z) + az^{-1}P(z) \Rightarrow \frac{P(z)}{X(z)} = \frac{1}{1-az^{-1}}$$

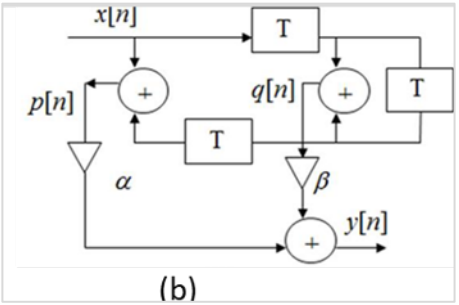
Also, $Y(z) = bP(z) + cz^{-1}P(z) \Rightarrow \frac{Y(z)}{P(z)} = b + cz^{-1}$

Therefore, $\frac{Y(z)}{X(z)} = \frac{b+cz^{-1}}{1-az^{-1}}$

$$Y(z)(1-az^{-1}) = X(z)(b+cz^{-1})$$

$$Y(z) = bX(z) + cz^{-1}X(z) + az^{-1}Y(z)$$

Hence, $\boxed{y[n] = bx[n] + cx[n-1] + ay[n-1]}$



(b)

$$q[n] = x[n-1] + x[n-2]; \qquad p[n] = x[n] + x[n-3]$$

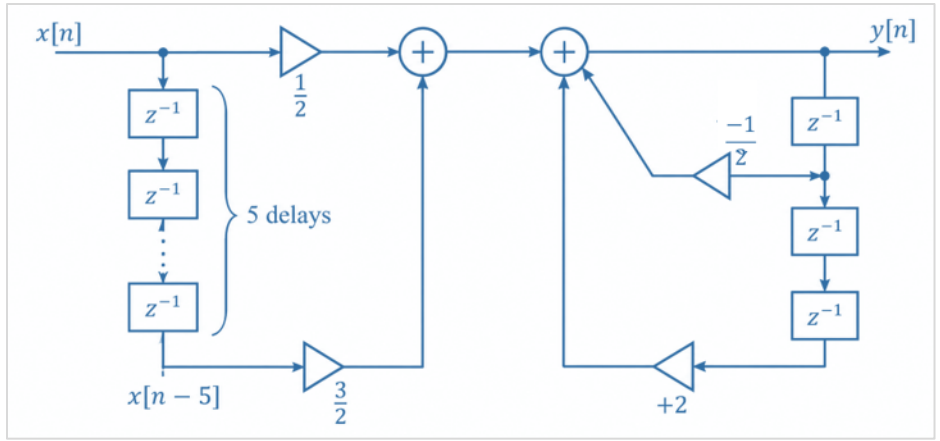
$$y[n] = \alpha p[n] + \beta q[n] = \alpha\{x[n] + x[n-3]\} + \beta\{x[n-1] + x[n-2]\}$$

Therefore, $\boxed{y[n] = \alpha x[n] + \beta x[n-1] + \beta x[n-2] + \alpha x[n-3]}$

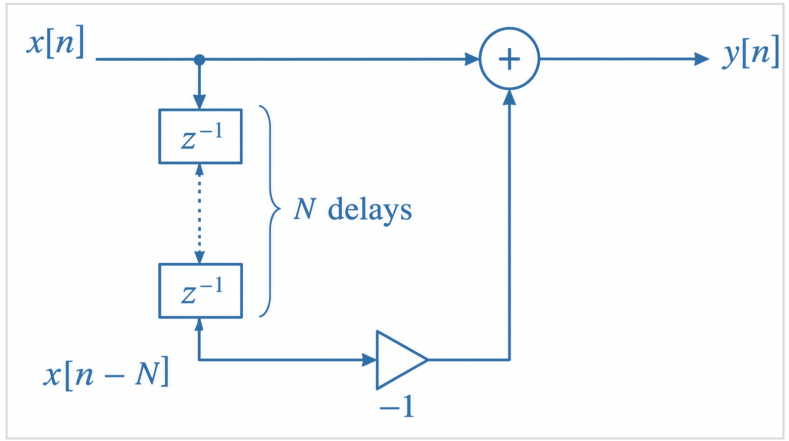
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3. Draw a system implementation for each of the following equations:

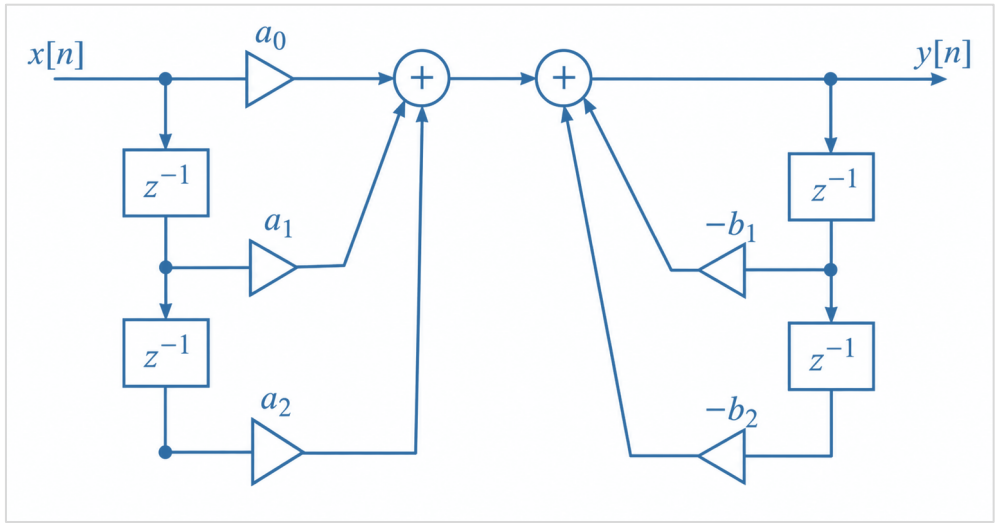
a) $2y[n] + y[n-1] - 4y[n-3] = x[n] + 3x[n-5]$



b) $y[n] = x[n] - x[n-N]$



c) $y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] - b_1y[n-1] - b_2y[n-2]$



4. A difference equation of a particular filter is given by

$$y[n] = 0.1x[n] - 0.1x[n-2] + 0.8x[n-3] + 0.1x[n-4] + 0.6x[n-6]$$

Find the impulse response of the above filter.

This is an FIR filter. $\therefore\ h[n] = \{\,, 0.1,\ ,0,\ ,-, 0.1\ \ 0.6\ \ 0.1\ \ 0\ \ 0.6\,\}$

with the first element corresponding to $n = 0$, and $h[n] = 0, n \geq 7$.

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5. Consider the stable causal system described by the difference equation $y[n] = ay[n-1] + bx[n]$ |a|<1

Let $x[n] = \delta[n]$ then $y[n] = h[n]$

For $n = 0$, $y[0] = ay[-1] + bx[0] = b$

For $n = 1$, $y[1] = ay[0] + bx[1] = ab$

For $n = 2$, $y[2] = ay[1] + bx[2] = a^2 b$

$\vdots$

For $n = n-1$, $y[n-1] = a^{n-1} b$

Therefore, $\boxed{y[n] = h[n] = a^n b, n \geq 0}$

Determine '*b*' in terms of '*a*' so that $\sum_{n=-\infty}^{\infty} h[n] = 1$.

$$\sum_{n=-\infty}^{\infty} h[n] = \sum_{n=0}^{\infty} a^n b = b[1 + a + a^2 + \cdots] = b\,\frac{1}{1-a}$$

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

Given that $\sum_{n=-\infty}^{\infty} h[n] = 1$,

therefore, $\frac{b}{1-a} = 1$

hence, $\boxed{b = 1 - a}$

6. Let $x[n]$ = {1 4 0 2} and $h[n]$ = {1 2 1}. Find their convolution (Both sequences start at $n = 0$).

$$X(z) = 1 + 4z^{-1} + 2z^{-3}; \qquad H(z) = 1 + 2z^{-1} + z^{-2}$$

$$Y(z) = X(z).H(z) = (1 + 4z^{-1} + 2z^{-3})(1 + 2z^{-1} + z^{-2})$$

$$= 1 + 2z^{-1} + z^{-2} + 4z^{-1} + 8z^{-2} + 4z^{-3} + 2z^{-3} + 4z^{-4} + 2z^{-5}$$

Therefore, $Y(z) = 1 + 6z^{-1} + 9z^{-2} + 6z^{-3} + 4z^{-4} + 2z^{-5}$

Taking the inverse z-transform,

$$\boxed{y[n] = x[n] * h[n] = \{1, \ , 6, \ , 9, \ 6\ 4\ 2\}}$$

7. Find the impulse response $h[n]$ for the following causal system:

$$H(z) = \frac{1 - \frac{1}{2}z^{-1}}{\left[1 - \frac{1}{4}z^{-1}\right]\left[1 + \frac{1}{4}z^{-1}\right]}$$

$$\frac{1 - \frac{1}{2}z^{-1}}{\left(1 - \frac{1}{4}z^{-1}\right)\left(1 + \frac{1}{4}z^{-1}\right)} = \frac{A}{1 - \frac{1}{4}z^{-1}} + \frac{B}{1 + \frac{1}{4}z^{-1}}$$

To find A, substitute $z^{-1} = 4$ (i.e. $1 - \frac{1}{4}z^{-1} = 0$); $A = \left.\frac{1-\frac{1}{2}z^{-1}}{1+\frac{1}{4}z^{-1}}\right|_{z^{-1}=4} = \frac{1-\frac{1}{2}(4)}{1+\frac{1}{4}(4)} = -\frac{1}{2}$

Similarly, $B = \left.\frac{1-\frac{1}{2}z^{-1}}{1-\frac{1}{4}z^{-1}}\right|_{z^{-1}=-4} = \frac{1-\frac{1}{2}(-4)}{1-\frac{1}{4}(-4)} = \frac{3}{2}$

Therefore, $H(z) = \frac{-\frac{1}{2}}{1-\frac{1}{4}z^{-1}} + \frac{\frac{3}{2}}{1+\frac{1}{4}z^{-1}}$; Hence, $\boxed{h[n] = -\frac{1}{2}\left(\frac{1}{4}\right)^n u[n] + \frac{3}{2}\left(-\frac{1}{4}\right)^n u[n]}$

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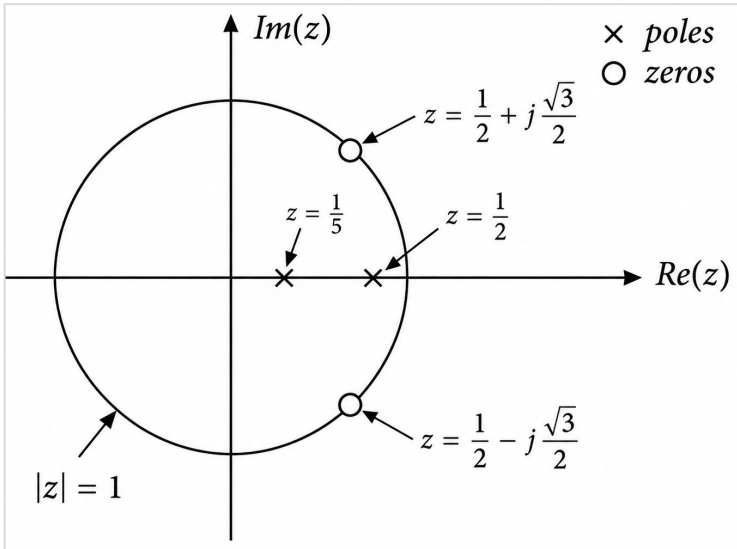
8. Consider the following system: $H(z) = \frac{1-2z^{-1}+2z^{-2}-z^{-3}}{(1-z^{-1})(1-0.5z^{-1})(1-0.2z^{-1})}$; (a) Sketch the pole-zero pattern. Is this system stable?

$$H(z) = \frac{1-2z^{-1}+2z^{-2}-z^{-3}}{(1-z^{-1})(1-0.5z^{-1})(1-0.2z^{-1})} = \frac{(1-z^{-1}+z^{-2})(1-z^{-1})}{(1-z^{-1})(1-0.5z^{-1})(1-0.2z^{-1})} = \frac{1-z^{-1}+z^{-2}}{\left(1-\frac{1}{2}z^{-1}\right)\left(1-\frac{1}{5}z^{-1}\right)}$$

The numerator can also be written as $1-z^{-1}+z^{-2} = \frac{z^2-z+1}{z^2}$;

Therefore, the zeros are obtained from $z^2-z+1=0$ giving $\boxed{z = \frac{1\pm j\sqrt{3}}{2}}$ 2 poles

Pole-zero plot



(b) Determine the impulse response of this causal system.

$H(z) = 1 + \frac{A}{1-\frac{1}{2}z^{-1}} + \frac{B}{1-\frac{1}{5}z^{-1}}$; Similar to Q7, we can find $A = \frac{5}{2}, B = -\frac{14}{5}$

Therefore, $H(z) = 1 + \frac{\left(\frac{5}{2}\right)z^{-1}}{1-\frac{1}{2}z^{-1}} - \frac{\left(\frac{14}{5}\right)z^{-1}}{1-\frac{1}{5}z^{-1}}$;

Taking the inverse z-transform, $h[n] = \delta[n] + \frac{5}{2}\left(\frac{1}{2}\right)^{n-1}u[n-1] - \frac{14}{5}\left(\frac{1}{5}\right)^{n-1}u[n-1]$

or equivalently, $\boxed{h[n] = \delta[n] + 5\left(\frac{1}{2}\right)^{n}u[n-1] - 14\left(\frac{1}{5}\right)^{n}u[n-1]}$

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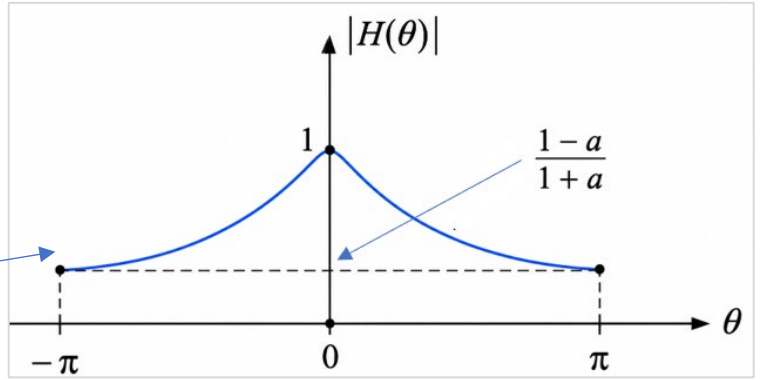
9. Determine and sketch the approximate magnitude response for the following filter: $H(z) = \frac{1-a}{1-az^{-1}}$ 0<a<1

$$H(\theta) = H(z)|_{z=e^{j\theta}} = \frac{1-a}{1-ae^{-j\theta}}$$

$$|H(\theta)|^2 = H(\theta)H^*(\theta) = \frac{1-a}{1-ae^{-j\theta}} \cdot \frac{1-a}{1-ae^{j\theta}} = \frac{(1-a)^2}{1-a(e^{j\theta}+e^{-j\theta})+a^2}$$

Since $e^{j\theta} + e^{-j\theta} = 2\cos\theta$, $\boxed{|H(\theta)|^2 = \frac{(1-a)^2}{1-2a\cos\theta+a^2}}$

$$\therefore \boxed{|H(\theta)| = \frac{1-a}{\sqrt{1-2a\cos\theta+a^2}}}$$



9. A speech signal is sampled with a sampling period of 125μs. A frame of 256 samples is selected and a 256 point DFT is computed. What is the spacing between the DFT values in Hz.

$$f_s = \frac{1}{125 \times 10^{-3}}$$

$$N = 256$$

$$\Delta f = \frac{f_s}{N} = \frac{1}{125 \times 10^{-3}} \times \frac{1}{256} = 0.03125\ Hz$$

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11. Let x[n]={1, 2, 1, 0}. With N=4, compute the 4-point DFT.

$X[k] = \sum_{n=0}^{N-1} x\,[n]e^{-j\frac{2\pi}{N}nk}$; Since $N = 4$, $X[k] = \sum_{n=0}^{3} x\,[n]e^{-j\frac{2\pi}{4}nk}$

$$X[k] = x[0] + x[1]e^{-j\frac{\pi}{2}k} + x[2]e^{-j\pi k} + x[3]e^{-j\frac{3\pi}{2}k}$$

Since $x[0] = 1,\ x[1] = 2,\ x[2] = 1,\ x[3] = 0,$

$$\boxed{X[k] = 1 + 2e^{-j\frac{\pi}{2}k} + e^{-j\pi k}}$$

For $k = 0,\ X[0] = 4$

For $k = 1,\ X[1] = 1 + 2e^{-j\frac{\pi}{2}} + e^{-j\pi} = -j2$

For $k = 2,\ X[2] = 1 + 2e^{-j\pi} + e^{-j2\pi} = 0$

For $k = 3,\ X[3] = 1 + 2e^{-j\frac{3\pi}{2}} + e^{-j3\pi} = j2$

Therefore, $X[k] = \{4, -j2, 0, j2\}$

12. Find the 8-point DFT of $x[n] = \{1, 1, 0, 0, 0, 0, 0, 0\}$, using conjugate symmetry property of $X[k]$.

Since $x[2]$ to $x[7]$ are all zeros, the upper index in the DFT summation will be $n = 1$.

$$X[k] = \sum_{n=0}^{1} x\,[n]e^{-j\frac{2\pi}{8}nk};\quad X[k] = x[0] + x[1]e^{-j\frac{2\pi}{8}k}$$

$$X[k] = 1 + e^{-j\frac{\pi}{4}k}, k = 0,1,2,\dots,7$$

Therefore, $X[0] = 1 + 1 = 2;\ X[1] = 1 + e^{-j\frac{\pi}{4}} = 1.707 - j0.707$

$$X[2] = 1 + e^{-j\frac{\pi}{2}} = 1 - j;\ X[3] = 1 + e^{-j\frac{3\pi}{4}} = 0.293 - j0.707$$

$$X[4] = 1 + e^{-j\pi} = 0$$

Using the conjugate symmetry property,

$$X[5] = X^*[3] = 0.293 + j0.707;\quad X[6] = X^*[2] = 1 + j$$

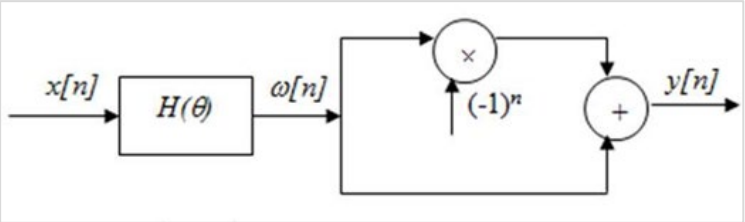
$$X[7] = X^*[1] = 1.707 + j0.707$$

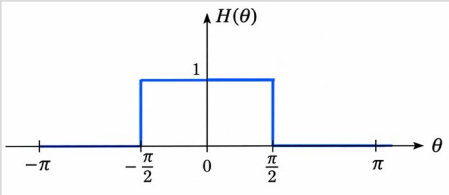
$X[k] =\ X[0],\ X[1],\ X[2],\ X[3],\ X[4],\ X[5],\ X[6]\ X[7]$

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13. For the system in the figure below, sketch the output *y[n]* when the input *x[n]* is $\delta[n]$ and $H(\theta)$ is an ideal lowpass filter as follows:

$$H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < |\theta| \le \pi \end{cases}$$





$$h[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\theta)e^{j\theta n}\,d\theta = \frac{1}{2\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2\,e^{j\theta n}\,d\theta;$$

$$h[n] = \frac{1}{2\pi}\frac{e^{j\frac{\pi}{2}n} - e^{-j\frac{\pi}{2}n}}{jn} = \frac{1}{\pi}\frac{\sin\left(\frac{n\pi}{2}\right)}{n}$$

Also, $y[n] = (-1)^n h[n] + h[n] = h[n]\{1 + (-1)^n\}$

Therefore, $y[n] = \frac{1}{2}\frac{\sin\left(\frac{n\pi}{2}\right)}{\frac{n\pi}{2}}\{1 + (-1)^n\}$

Hence, $\boxed{y[n] = \frac{1}{2}\{1 + (-1)^n\}\frac{\sin\left(\frac{n\pi}{2}\right)}{\frac{n\pi}{2}}}$

14. A 3- second signal is sampled at f_s = 100 Hz. The maximum spectral spacing is to be $\Delta f = 0.25\ Hz$. How many samples are needed for the DFT and FFT ?

Number of samples: $N = \frac{f_s}{\Delta f} = \frac{100}{0.25} = 400$

A 3-second signal gives 300 samples. To make 400 samples, we must pad the signal with 100 zeros. Therefore, for a 400-point for DFT.

For FFT: 512 *samples* $(2^9$ *is another option*$)$

i.e., 112 zeros must be added to obtain 512 samples.

The frequency spacing is $\Delta f = \frac{100}{512}$ 0.1953 *Hz.*

(better than the required frequency spacing of 0.25 Hz.)

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15. Classify the following system in terms of its linearity, time invariance, memory and causality: $y[n] = |x[-n]|$

Answer: *Non-linear, time invariant, dynamic, non-causal.*

1. Linearity

$$x_1[n] \xrightarrow{H} y_1[n] = |\, x_1[-n] \,|$$

$$x_2[n] \xrightarrow{H} y_2[n] = |\, x_2[-n] \,|$$

$$a_1 y_1[n] + a_2 y_2[n] = a_1 \,|\, x_1[-n] \,| + a_2 \,|\, x_2[-n] \,| \qquad (1)$$

whereas, $a_1 x_1[n] + a_2 x_2[n] \xrightarrow{H} |\, a_1 x_1[-n] + a_2 x_2[-n] \,|$ *(2)*

(1) and (2) are not the same. $\therefore$ *Non–linear*

2. Time invariance

$$x[n - m_0] \xrightarrow{H} |\, x[-(n - m_0)] \,| \qquad (3)$$

$$y[n - m_0] = |\, x[-(n - m_0)] \,| \qquad (4)$$

(3) and (4) are the same. $\therefore$ *Time invariant*

3. Memory

If $n = 2$, $y[2] = |\, x[-2] \,|$

The output relies on a past input. $\therefore$ *Dynamic*

4. Causal

If $n = -2$, $y[-2] = |\, x[2] \,|$

The output relies on a future input. $\therefore$ *Non–causal*

16. Consider a system with an input x[n] and an output y[n] which satisfies the following difference equation:

$$y[n] = n\, y[n-1] + x[n]$$

If $x[n] = \delta[n]$, determine $y[n]$ for all values of n.

$$x[n] = \delta[n]$$

For $n = 0$, $y[0] = 0\, y[-1] + x[0] = 1$

For $n = 1$, $y[1] = 1\, y[0] + x[1] = 1 \cdot 1$

For $n = 2$, $y[2] = 2\, y[1] + x[2] = 2 \cdot 1$

For $n = 3$, $y[3] = 3\, y[2] + x[3] = 3 \cdot 2 \cdot 1$

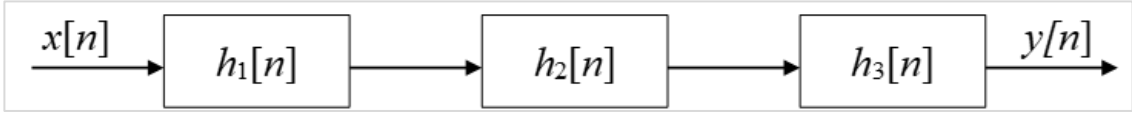
For $n = 4$, $y[4] = 4\, y[3] + x[4] = 4 \cdot 3 \cdot 2 \cdot 1$

Therefore, for general n, $y[n] = n(n-1)(n-2)(n-3)\cdots 1 = n!$

Hence, $\boxed{y[n] = n!, n \geq 1}$

Topic 2: Pass – Level Tutorial Questions

17. Three systems with the following impulse responses are connected in cascade: $h_1[\mathrm{n}] = \delta(n) - \delta(n-1)$; $h_2[\mathrm{n}] = \mathrm{g}[\mathrm{n}]$; $h_3[\mathrm{n}] = u[n]$; where $u[n]$ is the unit step sequence and $\delta[n]$ is the unit sample sequence (i.e an impulse).



What is the impulse response $h_c(n)$ of the overall system?

$h_1[n] = \delta[n] - \delta[n-1]$; *Therefore,* $H_1(z) = 1 - z^{-1}$

$h_3[n] = u[n]$; *Therefore,* $H_3(z) = \frac{1}{1-z^{-1}}$

$$y[n] = x[n] * h_1[n] * h_2[n] * h_3[n]$$

$$\frac{Y(z)}{X(z)} = H_1(z) \cdot H_2(z) \cdot H_3(z)$$

$$H_c(z) = \left(1 - z^{-1}\right) \cdot G(z) \cdot \frac{1}{1 - z^{-1}}$$

Therefore, $H_c(z) = G(z)$

Hence, $\boxed{h_c[n] = g[n]}$

18. Find the z-transform, G(z) and its ROC , if $g[n] = \begin{cases} na^n & n \geq 0 \\ 0 & n < 0 \end{cases}$

where *a* is a constant

Let $x[n] = a^n$; *then,* $X(z) = \frac{z}{z-a}$

$$\therefore \frac{dX(z)}{dz} = \frac{-a}{(z-a)^2}$$

Given $g[n] = na^n$

using the z-transform property,

$$G(z) = -z\frac{dX(z)}{dz}$$

Therefore, $G(z) = -z\left[\frac{-a}{(z-a)^2}\right] = \frac{az}{(z-a)^2}$

$$= \frac{az}{z^2(1-az^{-1})^2} = \frac{az^{-1}}{(1-az^{-1})^2}$$

Hence,

$$\boxed{G(z) = \frac{az^{-1}}{(1-az^{-1})^2}}$$

Topic 2: Pass – Level Tutorial Questions

19. Consider the interconnection of LTI systems as shown in Figure 2:

(a) Express the overall impulse response in terms of $h_1[n]$, $h_2[n]$, $h_3[n]$ and $h_4[n]$.

$$\boxed{h[n] = h_1[n] * \left[h_2[n] - h_3[n] * h_4[n]\right]}$$

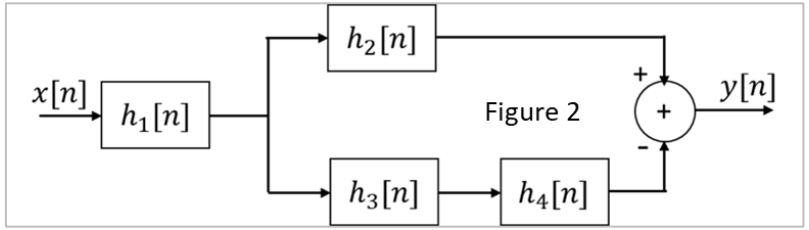


Figure 2

(b) Determine *h[n]* when $h_1[n] = \{\frac{1}{2} \;\; \frac{1}{4} \;\; \frac{1}{2}\}$; $h_2[n] = h_3[n] = (n+1)\,u[n]$; $h_4[n] = \delta[n-2]$

$$h_3[n] * h_4[n] = (n+1)u[n] * \delta[n-2] = (n-1)u[n-2]$$

Therefore, $h_2[n] - h_3[n] * h_4[n] = (n+1)u[n] - (n-1)u[n-2]$

$$= n\{u[n] - u[n-2]\} + \{u[n] + u[n-2]\} = 2u[n] - \delta[n]$$

Given in (b), $h_1[n] = \frac{1}{2}\delta[n] + \frac{1}{4}\delta[n-1] + \frac{1}{2}\delta[n-2]$

Hence, $h[n] = \left[\frac{1}{2}\delta[n] + \frac{5}{4}\delta[n-1] + \frac{1}{2}\delta[n-2]\right] * \left[2u[n] - \delta[n]\right]$

Expanding gives $h[n] = \mathrm{u}[n] + \frac{1}{2}u[n-1] + u[n-2] - \frac{1}{2}\delta[n] - \frac{1}{4}[n-1] - \frac{1}{2}\delta[n-2]$

Therefore, $\boxed{h[n] = \frac{1}{2}\delta[n] + \frac{5}{4}\delta[n-1] + 2\delta[n-2] + \frac{5}{2}u[n-3]}$

The $h[n]$ could be written entirely using $u[n], u[n-1], u[n-2], u[n-3]$

$$h[n] = \frac{1}{2}u[n] + \frac{3}{4}u[n-1] + \frac{3}{4}u[n-2] + \frac{1}{2}u[n-3]$$

Note: $H(z) = \left[\frac{1}{2} + \frac{1}{4}z^{-1} + \frac{1}{2}z^{-2}\right]\left[\frac{2}{1-z^{-1}} - 1\right]$

$$= \frac{\frac{1}{2} + \frac{3}{4}z^{-1} + \frac{3}{4}z^{-2} + \frac{1}{2}z^{-3}}{1 - z^{-1}}$$

Taking inverse z-Transform , we obtain:

$$\boxed{h[n] = \frac{1}{2}u[n] + \frac{3}{4}u[n-1] + \frac{3}{4}u[n-2] + \frac{1}{2}u[n-3]}$$

(c) Determine the output $y[n]$ of the system in part(b) for the input $x[n] = \delta[n+2] + 3\delta[n-1] - 4\delta[n-3]$

$$\begin{cases} n = 0 \\ x[n] = \{1, 0, \underset{\downarrow}{0}, 3, 0\ -4\}, \\ h[n] = \frac{1}{2}\delta[n] + \frac{5}{4}\delta[n-1] + 2\delta[n-2] + \frac{5}{2}u[n-3] \end{cases}$$

$y[n] = x[n] * h[n] = \{\frac{1}{2},\ \frac{5}{4},\ 2,\ \frac{25}{4},\ \frac{13}{2},\ 5,\ 2,\ 0, 0 \ldots.\}$

$n = 0$